EXPERIMENT 02A

Medical GPT Baseline Training Report

Medical Domain Corpus | ~5 Million Tokens | GPT-2 BPE Tokenizer

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Dataset	Hardware	Date	Epochs
Medical Corpus (~5M tokens)	Google Colab T4	May 2026	5

1. Objective

Experiment 02A established a stable medical-domain GPT training baseline using a larger medical corpus and the GPT-2 BPE tokenizer. The experiment focused on improving generalization over Experiment 01, increasing model capacity, reducing training-window overlap, and preserving the best checkpoints during training.

2. Model Configuration

The model architecture follows a standard decoder-only transformer design. Relative to Experiment 01, the model capacity was scaled substantially to better match the larger medical-domain training corpus. This configuration produced approximately 70 million parameters.

Component	Configuration	Parameters (Approx.)
Token Embedding	50,257 × 512	25.7 M
Position Embedding	256 × 512	0.13 M
Transformer Layers	6 Layers | d_model=512 | heads=8	~18–20 M
LM Head	50,257 × 512 | Untied	25.7 M
Total	Decoder-only Transformer	~70 M

3. Hyperparameter Configuration

Parameter	Value
Vocabulary Size	50,257
Context Length	256
d_model	512
Attention Heads	8
Transformer Layers	6
Dropout	0.1
Learning Rate	3e-4
Minimum Learning Rate	1e-5
Gradient Accumulation	8 Steps
Mixed Precision (AMP)	Enabled

4. Training Setup

Setting	Value
Optimizer	AdamW
Weight Decay	0.1
Gradient Clipping	1.0
Learning Rate Schedule	Cosine Decay
Sliding Window Stride	64 Tokens
Checkpoint Strategy	latest.pt + best.pt + epoch_N.pt
Flash Attention	Not Used

5. Training Results

Epoch	Train Loss	Validation Loss	Perplexity
1	5.4626	4.6626	105.9074
2	4.0017	4.2176	67.8692
3	3.5383	4.0732	58.7436
4	3.3000	4.0313	56.3350
5	3.1951	4.0208	55.7466

6. Analysis

Validation loss decreased consistently across all five epochs, while perplexity fell from 105.9074 to 55.7466. Unlike Experiment 01, where validation loss increased every epoch, Experiment 02A demonstrated stable generalization on the medical corpus.

7. Root Cause Analysis

Factor	Status	Observation
Dataset Size	Improved	Scaled from ~300K to ~5M tokens
Model Capacity	Scaled	Increased from ~16M to ~70M parameters
Sliding Window	Improved	Stride set to 64 tokens
Checkpoint Strategy	Fixed	best.pt and epoch checkpoints preserved
Regularization	Stable	Dropout remained at 0.1

8. Comparison with Experiment 01

Metric	Experiment 01	Experiment 02A
Dataset Size	~300K tokens	~5M tokens
Model Parameters	~16M	~70M
Domain	WikiText	Medical
Final Validation Loss	11.9652	4.0208

Final Perplexity	157,185	55.7466
Validation Trend	Increasing	Decreasing
Checkpoint Strategy	latest.pt only	best.pt + epoch checkpoints

9. Conclusion

Experiment 02A successfully established the first stable generalization baseline in the project series. Scaling the dataset to approximately 5 million medical-domain tokens and increasing model capacity to approximately 70 million parameters substantially improved validation behaviour compared to Experiment 01. The improved checkpoint strategy also preserved the best-performing model throughout training.

The final validation loss of 4.0208 and perplexity of 55.7466 provide the reference baseline for Experiment 02B, where the same architecture and medical corpus are evaluated using a custom biomedical BPE tokenizer.